Onboarding Snakemake: Progress Towards WfExS-backend Integration

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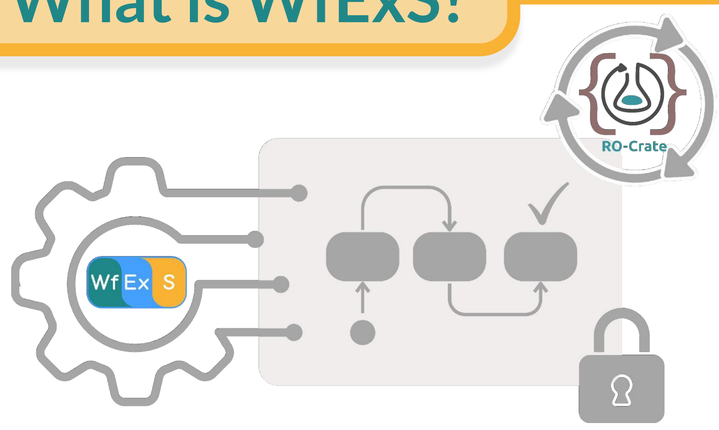
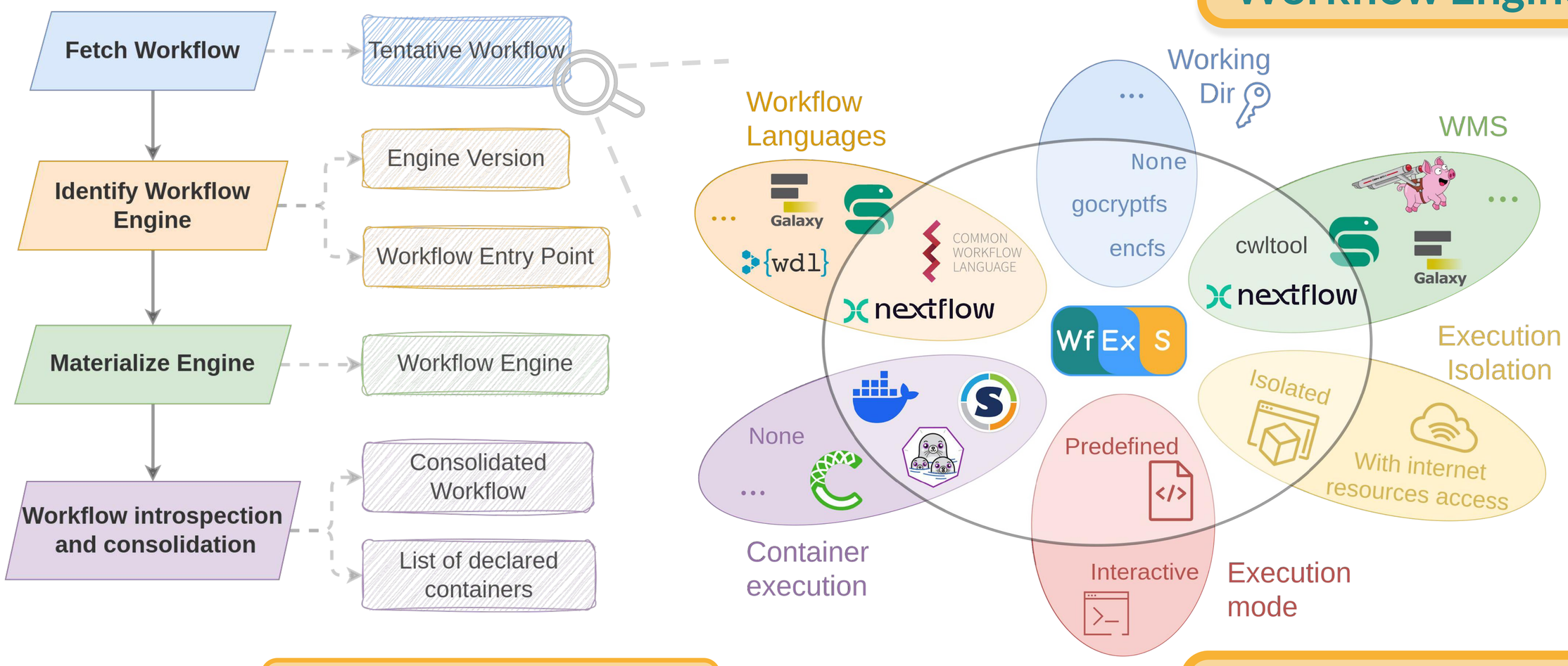
What is WfExS?



- High-level **workflow execution orchestrator**.
- **Secure** workflow executions in isolated environments using containers.
- Fetches and materialize all the elements needed to instantiate a workflow.
- **Reproducible** executions using Workflow Run RO-Crate as representation.

Scan me!

Workflow Engines

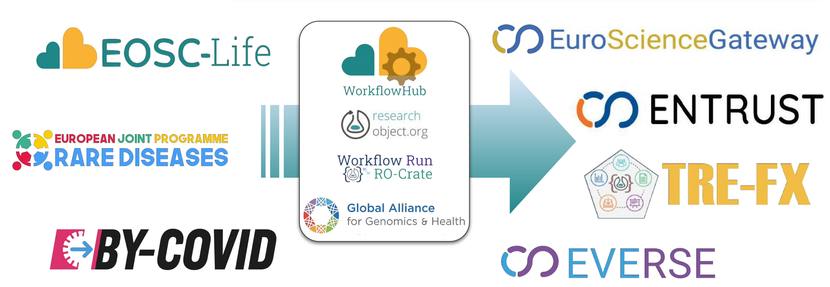


Onboarding Snakemake

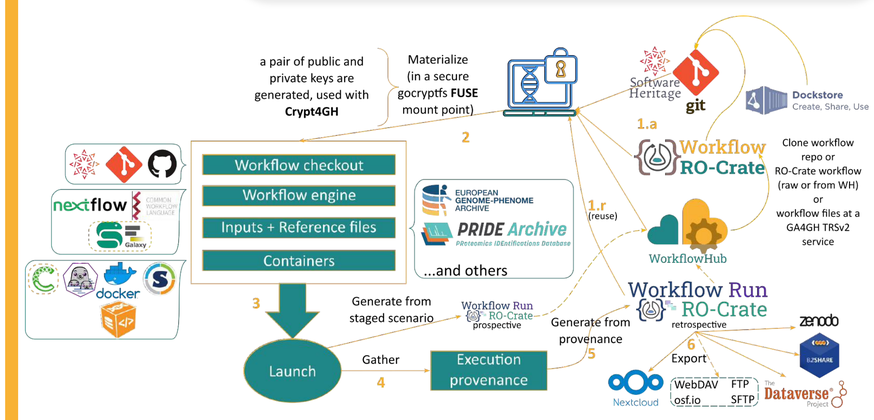
	CWL (COMMON WORKFLOW LANGUAGE)	Nextflow	Snakemake
Workflow source format	YAML	Java → Groovy	
Workflow language	CWL 1.0 , 1.1 and 1.2	Nextflow (v1 and v2)	Up to v8
Implementation	Multiple (cwltool, toil, cwlTES, cromwell < 80)	Single	Single
Introspection without workflow execution	✓	Partial (WfExS had to use a groovy parser)	Partial
Clear separation of inputs/outputs/params	✓	✗ (partially using nf-core tooling)	✗
Modularity	✓	v2 only	✓
Workflow elements locality	Local allowing remote references	Local only (remote references management using nf-core tooling)	Local allowing remote references (wrappers, modules, meta-wrappers)
Consolidate workflow without execution	✓	N/A (remote references materialization using nf-core tooling)	✗
Materialize containers without execution	✗	✗	✗

Complexity: Moderate → Challenging → Complex →

Past and present projects



WfExS Context



wfexs.community-join@bsc.es

Ask joining WfExS general channel

Documentation

Github

Contact & Funding

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